

Keshav Malhotra

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EDUCATION

University of Waterloo

BASc, Mechatronics Engineering (Co-op)

2025 - Present

Waterloo, ON

SKILLS

Programming: C++, Python, Java, JavaScript, HTML/CSS, SQL

Machine Learning: PyTorch, OpenCV, pandas, NumPy, pydicom, YOLO

EXPERIENCE

The Hospital for Sick Children (SickKids) - PCIGITI/Neurovascular Lab May 2026 – Aug. 2026

ML Research Intern

Toronto, ON

- Cut contrast-dye detection latency from **~350 ms to ~50 ms (7×)** in fluoroscopy (DICOM X-ray) angiography by engineering **AI Trace**, an end-to-end real-time detection pipeline in **Python** (PyTorch, NumPy, OpenCV) sustaining **~13 FPS** with **84% of frames under 100 ms** across a 1,142-frame benchmark.
- Overcame the absence of labeled patient data by training a **PyTorch** ResNet34 U-Net on only **19 annotated simulated frames**, generalizing to a held-out real-patient cerebral AVM study at **0.80 Dice / 0.88 recall** with zero patient data in training.
- Eliminated manual ROI selection by designing an automatic-ROI feature on a ResNet34 U-Net, reaching **0.82 mean held-out recall** on fine-vessel segmentation, **2×** the 0.41 recall of the prior **YOLO** prototype.

The Hospital for Sick Children (SickKids) - AI/ED Lab

Jul. 2024 – Oct. 2024

High School Research Intern

Toronto, ON

- Shaped the conversational design of **SKAI (SickKids AI)**, the hospital's first patient-facing AI chatbot, translating qualitative findings from **100+ patient interviews** into LLM tuning recommendations on temperature and sentiment alignment, adopted in the deployed configuration to give worried parents clear, concise answers before seeing a physician.
- Prototyped a mobile autonomous robot for emergency department logistics, building ultrasonic-sensor navigation and Arduino-based control to move supplies through high-traffic clinical corridors.

FRC - Team 6141

Jun. 2024 – Jun. 2025

School Captain

Toronto, ON

- Developed a **Java-based control system** for a differential drive robot using WPILib, implementing Arcade Drive logic for intuitive teleoperated handling via Xbox controller.
- Mentored **30+ students** in **Git version control**, object-oriented programming, and sensor integration (ultrasonic/IR), establishing a standardized documentation pipeline.

PROJECTS

JobFit: AI Resume Analyzer ▶ Demo | Python, Flask, OpenAI, NumPy, JavaScript

Mar. 2026

- Engineered a **RAG-powered** resume analyzer using **Flask** and **GPT-3.5-turbo**, implementing **vector embeddings** and **cosine similarity** to semantically match resume chunks against job postings.
- Designed a **JSON-structured** prompt pipeline with input validation to minimize hallucinations, returning an ATS match score, missing keywords, and feedback via an interactive frontend.

NavBot - Autonomous Hospital Navigation Robot ▶ Demo | C++, Git

Nov. 2025 – Dec. 2025

- Developed **autonomous navigation software** for a hospital delivery robot, handling real-time decision-making in dynamic indoor environments.
- Implemented **Dijkstra's algorithm** to calculate optimal global paths in a static map, integrated with real-time local path planning to navigate dynamic hospital environments.
- Experimentally tuned a custom PI controller for differential drive motors, achieving rotational accuracy within **1 degree error** and **linear precision within 5cm** using bumper and ultrasonic sensor feedback.

Roboventure ▶ Demo | Arduino, Git, Excel

Aug. 2024 – Sept. 2024

- Spearheaded technical workshops for **50+ students**, delivering hands-on instruction in **Arduino** firmware development, circuit prototyping, and electrical safety practices.
- Scaled organization participation by **300%** within one month by architecting a targeted outreach strategy and executing high-engagement robotics demonstrations.